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1 "D:\Poojangi\Research Paper\GA for TSP\venv\Scripts\python.exe" "D:/Poojangi/
  Research Paper/GA for TSP/testing.py"
2 Iteration No: 0
3 The best route so far is: [0, 12, 4, 6, 8, 7, 5, 11, 2, 1, 14, 9, 13, 10, 3]
  with distance: 491.0
4 -----
5 Iteration No: 1
6 The best route so far is: [0, 12, 4, 5, 11, 2, 7, 8, 6, 1, 14, 9, 13, 10, 3]
  with distance: 550.0
7 -----
8 Iteration No: 2
9 The best route so far is: [13, 9, 2, 4, 5, 11, 7, 8, 6, 1, 14, 10, 3, 12, 0]
  with distance: 518.0
10 -----
11 Iteration No: 3
12 The best route so far is: [13, 9, 2, 4, 5, 11, 7, 8, 6, 1, 14, 10, 3, 12, 0]
  with distance: 518.0
13 -----
14 Iteration No: 4
15 The best route so far is: [13, 9, 2, 4, 5, 11, 7, 8, 6, 1, 14, 10, 3, 12, 0]
  with distance: 518.0
16 -----
17 Iteration No: 5
18 The best route so far is: [13, 9, 2, 4, 5, 11, 7, 8, 6, 1, 14, 10, 3, 12, 0]
  with distance: 518.0
19 -----
20 Iteration No: 6
21 The best route so far is: [13, 9, 2, 4, 5, 11, 7, 8, 6, 1, 14, 10, 3, 12, 0]
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21 with distance: 518.0
22 -----
23 Iteration No: 7
24 The best route so far is: [0, 12, 7, 8, 1, 14, 9, 11, 5, 6, 4, 2, 13, 10, 3]
25 with distance: 515.0
26 -----
27 Iteration No: 8
28 The best route so far is: [0, 12, 7, 8, 1, 14, 9, 11, 5, 6, 4, 2, 13, 10, 3]
29 with distance: 515.0
30 -----
31 Iteration No: 9
32 The best route so far is: [0, 12, 7, 8, 1, 14, 9, 11, 5, 6, 4, 2, 13, 10, 3]
33 with distance: 515.0
34 -----
35 Iteration No: 10
36 The best route so far is: [0, 12, 7, 8, 1, 14, 9, 11, 5, 6, 4, 2, 13, 10, 3]
37 with distance: 515.0
38 -----
39 Iteration No: 11
40 The best route so far is: [0, 12, 7, 8, 1, 14, 9, 11, 5, 6, 4, 2, 13, 10, 3]
41 with distance: 515.0
42 -----
43 Iteration No: 12
44 The best route so far is: [0, 12, 7, 8, 1, 14, 9, 11, 5, 6, 4, 2, 13, 10, 3]
45 with distance: 515.0
46 -----
47 Iteration No: 13
48 The best route so far is: [0, 12, 7, 8, 1, 14, 9, 11, 5, 6, 4, 2, 13, 10, 3]
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42 with distance: 515.0
43 -----
44 Iteration No: 14
45 The best route so far is: [0, 12, 7, 8, 1, 14, 9, 11, 5, 6, 4, 2, 13, 10, 3]
46 with distance: 515.0
47 -----
48 Iteration No: 15
49 The best route so far is: [7, 14, 11, 8, 5, 9, 4, 6, 2, 13, 3, 10, 1, 12, 0]
50 with distance: 486.0
51 -----
52 Iteration No: 16
53 The best route so far is: [7, 14, 11, 8, 5, 9, 4, 6, 2, 13, 3, 10, 1, 12, 0]
54 with distance: 486.0
55 -----
56 Iteration No: 17
57 The best route so far is: [7, 14, 11, 8, 5, 9, 4, 6, 2, 13, 3, 10, 1, 12, 0]
58 with distance: 486.0
59 -----
60 Iteration No: 18
61 The best route so far is: [7, 14, 11, 8, 5, 9, 4, 6, 2, 13, 3, 10, 1, 12, 0]
62 with distance: 486.0
63 -----
64 Iteration No: 19
65 The best route so far is: [7, 14, 11, 8, 5, 9, 4, 6, 2, 13, 3, 10, 1, 12, 0]
66 with distance: 486.0
67 -----
68 Iteration No: 20
69 The best route so far is: [14, 12, 1, 7, 5, 9, 4, 6, 2, 13, 3, 10, 11, 8, 0]
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63 With distance: 460.0
64 -----
65 Iteration No: 21
66 The best route so far is: [14, 12, 1, 7, 5, 9, 4, 6, 2, 13, 3, 10, 11, 8, 0]
67 with distance: 460.0
68 -----
69 Iteration No: 22
69 The best route so far is: [14, 7, 5, 9, 11, 8, 4, 6, 2, 13, 3, 10, 1, 12, 0]
70 with distance: 395.0
71 -----
71 Iteration No: 23
72 The best route so far is: [14, 12, 1, 7, 5, 9, 4, 6, 2, 13, 3, 10, 11, 8, 0]
73 with distance: 460.0
74 -----
74 Iteration No: 24
75 The best route so far is: [14, 12, 1, 7, 5, 9, 4, 6, 2, 13, 3, 10, 11, 8, 0]
76 with distance: 460.0
77 -----
77 Iteration No: 25
78 The best route so far is: [14, 12, 1, 7, 5, 9, 4, 6, 2, 13, 3, 10, 11, 8, 0]
79 with distance: 460.0
80 -----
80 Iteration No: 26
81 The best route so far is: [14, 12, 1, 7, 5, 9, 4, 6, 2, 13, 3, 10, 11, 8, 0]
82 with distance: 460.0
83 -----
83 Iteration No: 27
84 The best route so far is: [14, 0, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 12]
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84 with distance: 481.0
85 -----
86 Iteration No: 28
87 The best route so far is: [14, 0, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 12]
88 with distance: 481.0
89 -----
89 Iteration No: 29
90 The best route so far is: [14, 0, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 12]
91 with distance: 481.0
92 -----
92 Iteration No: 30
93 The best route so far is: [14, 0, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 12]
94 with distance: 481.0
95 -----
95 Iteration No: 31
96 The best route so far is: [14, 0, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 12]
97 with distance: 481.0
98 -----
98 Iteration No: 32
99 The best route so far is: [14, 0, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 12]
100 with distance: 481.0
101 -----
101 Iteration No: 33
102 The best route so far is: [14, 0, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 12]
103 with distance: 481.0
104 -----
104 Iteration No: 34
105 The best route so far is: [14, 0, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 12]
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105 with distance: 481.0
106 -----
107 Iteration No: 35
108 The best route so far is: [0, 14, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 12]
109 with distance: 441.0
110 -----
110 Iteration No: 36
111 The best route so far is: [0, 14, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 12]
112 with distance: 441.0
113 -----
113 Iteration No: 37
114 The best route so far is: [0, 14, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 12]
115 with distance: 441.0
116 -----
116 Iteration No: 38
117 The best route so far is: [14, 12, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 0]
118 with distance: 431.0
119 -----
119 Iteration No: 39
120 The best route so far is: [14, 12, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 0]
121 with distance: 431.0
122 -----
122 Iteration No: 40
123 The best route so far is: [14, 12, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 0]
124 with distance: 431.0
125 -----
125 Iteration No: 41
126 The best route so far is: [14, 12, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 0]
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126 with distance: 431.0
127 -----
128 Iteration No: 42
129 The best route so far is: [14, 12, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 0]
130 with distance: 431.0
131 -----
131 Iteration No: 43
132 The best route so far is: [14, 12, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 0]
133 with distance: 431.0
134 -----
134 Iteration No: 44
135 The best route so far is: [14, 12, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 0]
136 with distance: 431.0
137 -----
137 Iteration No: 45
138 The best route so far is: [14, 12, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 0]
139 with distance: 431.0
140 -----
140 Iteration No: 46
141 The best route so far is: [14, 12, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 0]
142 with distance: 431.0
143 -----
143 Iteration No: 47
144 The best route so far is: [14, 12, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 0]
145 with distance: 431.0
146 -----
146 Iteration No: 48
147 The best route so far is: [14, 12, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 0]
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147 with distance: 431.0
148 -----
149 Iteration No: 49
150 The best route so far is: [14, 12, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 0]
151 with distance: 431.0
152 -----
153 Iteration No: 50
154 The best route so far is: [14, 12, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 0]
155 with distance: 431.0
156 -----
157 Iteration No: 51
158 The best route so far is: [14, 12, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 0]
159 with distance: 431.0
160 -----
161 Iteration No: 52
162 The best route so far is: [14, 12, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 0]
163 with distance: 431.0
164 -----
165 Iteration No: 53
166 The best route so far is: [14, 12, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 0]
167 with distance: 431.0
168 -----
169 Iteration No: 54
170 The best route so far is: [14, 12, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 0]
171 with distance: 431.0
172 -----
173 Iteration No: 55
174 The best route so far is: [14, 12, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 0]
```

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168 with distance: 431.0
169 -----
170 Iteration No: 56
171 The best route so far is: [14, 12, 8, 9, 5, 3, 10, 11, 4, 6, 2, 13, 7, 1, 0]
172 with distance: 431.0
173 -----
174 Iteration No: 57
175 The best route so far is: [12, 9, 11, 2, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
176 with distance: 404.0
177 -----
178 Iteration No: 58
179 The best route so far is: [12, 9, 11, 2, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
180 with distance: 404.0
181 -----
182 Iteration No: 59
183 The best route so far is: [12, 9, 11, 2, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
184 with distance: 404.0
185 -----
186 Iteration No: 60
187 The best route so far is: [12, 9, 11, 2, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
188 with distance: 404.0
189 -----
190 Iteration No: 61
191 The best route so far is: [12, 9, 11, 2, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
192 with distance: 404.0
193 -----
194 Iteration No: 62
195 The best route so far is: [12, 9, 11, 2, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
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189 with distance: 404.0
190 -----
191 Iteration No: 63
192 The best route so far is: [12, 9, 11, 2, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
193 with distance: 404.0
194 -----
194 Iteration No: 64
195 The best route so far is: [12, 9, 11, 2, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
196 with distance: 404.0
197 -----
197 Iteration No: 65
198 The best route so far is: [12, 9, 11, 2, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
199 with distance: 404.0
200 -----
200 Iteration No: 66
201 The best route so far is: [12, 9, 11, 2, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
202 with distance: 404.0
203 -----
203 Iteration No: 67
204 The best route so far is: [12, 9, 11, 2, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
205 with distance: 404.0
206 -----
206 Iteration No: 68
207 The best route so far is: [12, 9, 11, 2, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
208 with distance: 404.0
209 -----
209 Iteration No: 69
210 The best route so far is: [12, 9, 11, 2, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
```

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210 with distance: 404.0
211 -----
212 Iteration No: 70
213 The best route so far is: [12, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
214 with distance: 392.0
215 -----
216 Iteration No: 71
217 The best route so far is: [12, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
218 with distance: 392.0
219 -----
220 Iteration No: 72
221 The best route so far is: [12, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
222 with distance: 392.0
223 -----
224 Iteration No: 73
225 The best route so far is: [12, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
226 with distance: 392.0
227 -----
228 Iteration No: 74
229 The best route so far is: [12, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
230 with distance: 392.0
231 -----
232 Iteration No: 75
233 The best route so far is: [12, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
234 with distance: 392.0
235 -----
236 Iteration No: 76
237 The best route so far is: [12, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
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231 with distance: 392.0
232 -----
233 Iteration No: 77
234 The best route so far is: [12, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
    with distance: 392.0
235 -----
236 Iteration No: 78
237 The best route so far is: [12, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
    with distance: 392.0
238 -----
239 Iteration No: 79
240 The best route so far is: [12, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
    with distance: 392.0
241 -----
242 Iteration No: 80
243 The best route so far is: [12, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
    with distance: 392.0
244 -----
245 Iteration No: 81
246 The best route so far is: [12, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
    with distance: 392.0
247 -----
248 Iteration No: 82
249 The best route so far is: [12, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
    with distance: 392.0
250 -----
251 Iteration No: 83
252 The best route so far is: [12, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
```

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252 with distance: 392.0
253 -----
254 Iteration No: 84
255 The best route so far is: [12, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 6, 1, 0]
    with distance: 392.0
256 -----
257 Iteration No: 85
258 The best route so far is: [6, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 12, 1, 0]
    with distance: 356.0
259 -----
260 Iteration No: 86
261 The best route so far is: [6, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 12, 1, 0]
    with distance: 356.0
262 -----
263 Iteration No: 87
264 The best route so far is: [6, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 12, 1, 0]
    with distance: 356.0
265 -----
266 Iteration No: 88
267 The best route so far is: [6, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 12, 1, 0]
    with distance: 356.0
268 -----
269 Iteration No: 89
270 The best route so far is: [6, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 12, 1, 0]
    with distance: 356.0
271 -----
272 Iteration No: 90
273 The best route so far is: [6, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 12, 1, 0]
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273 with distance: 356.0
274 -----
275 Iteration No: 91
276 The best route so far is: [6, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 12, 1, 0]
277 with distance: 356.0
278 -----
279 Iteration No: 92
280 The best route so far is: [6, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 12, 1, 0]
281 with distance: 356.0
282 -----
283 Iteration No: 93
284 The best route so far is: [6, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 12, 1, 0]
285 with distance: 356.0
286 -----
287 Iteration No: 94
288 The best route so far is: [6, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 12, 1, 0]
289 with distance: 356.0
290 -----
291 Iteration No: 95
292 The best route so far is: [6, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 12, 1, 0]
293 with distance: 356.0
294 -----
295 Iteration No: 96
296 The best route so far is: [6, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 12, 1, 0]
297 with distance: 356.0
298 -----
299 Iteration No: 97
300 The best route so far is: [6, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 12, 1, 0]
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294 with distance: 356.0
295 -----
296 Iteration No: 98
297 The best route so far is: [6, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 12, 1, 0]
298 with distance: 356.0
299 -----
299 Iteration No: 99
300 The best route so far is: [6, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 12, 1, 0]
301 with distance: 356.0
302 -----
302 Iteration No: 100
303 The best route so far is: [6, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 12, 1, 0]
304 with distance: 356.0
305 -----
305 Iteration No: 101
306 The best route so far is: [6, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 12, 1, 0]
307 with distance: 356.0
308 -----
308 Iteration No: 102
309 The best route so far is: [6, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 12, 1, 0]
310 with distance: 356.0
311 -----
311 Iteration No: 103
312 The best route so far is: [6, 9, 2, 11, 13, 7, 5, 3, 10, 14, 4, 8, 12, 1, 0]
313 with distance: 356.0
314 -----
314 Iteration No: 104
315 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
```

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315 with distance: 356.0
316 -----
317 Iteration No: 105
318 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
    with distance: 356.0
319 -----
320 Iteration No: 106
321 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
    with distance: 356.0
322 -----
323 Iteration No: 107
324 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
    with distance: 356.0
325 -----
326 Iteration No: 108
327 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
    with distance: 356.0
328 -----
329 Iteration No: 109
330 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
    with distance: 356.0
331 -----
332 Iteration No: 110
333 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
    with distance: 356.0
334 -----
335 Iteration No: 111
336 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
```

```
336 with distance: 356.0
337 -----
338 Iteration No: 112
339 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
340 with distance: 356.0
341 -----
342 Iteration No: 113
343 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
344 with distance: 356.0
345 -----
346 Iteration No: 114
347 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
348 with distance: 356.0
349 -----
350 Iteration No: 115
351 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
352 with distance: 356.0
353 -----
354 Iteration No: 116
355 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
356 with distance: 356.0
357 -----
358 Iteration No: 117
359 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
360 with distance: 356.0
361 -----
362 Iteration No: 118
363 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
```

```
357 with distance: 356.0
358 -----
359 Iteration No: 119
360 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
    with distance: 356.0
361 -----
362 Iteration No: 120
363 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
    with distance: 356.0
364 -----
365 Iteration No: 121
366 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
    with distance: 356.0
367 -----
368 Iteration No: 122
369 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
    with distance: 356.0
370 -----
371 Iteration No: 123
372 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
    with distance: 356.0
373 -----
374 Iteration No: 124
375 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
    with distance: 356.0
376 -----
377 Iteration No: 125
378 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
```

```
378 with distance: 356.0
379 -----
380 Iteration No: 126
381 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
382 with distance: 356.0
383 -----
383 Iteration No: 127
384 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
385 with distance: 356.0
386 -----
386 Iteration No: 128
387 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
388 with distance: 356.0
389 -----
389 Iteration No: 129
390 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
391 with distance: 356.0
392 -----
392 Iteration No: 130
393 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
394 with distance: 356.0
395 -----
395 Iteration No: 131
396 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
397 with distance: 356.0
398 -----
398 Iteration No: 132
399 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
```

```
399 with distance: 356.0
400 -----
401 Iteration No: 133
402 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
403 with distance: 356.0
404 -----
404 Iteration No: 134
405 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
406 with distance: 356.0
407 -----
407 Iteration No: 135
408 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
409 with distance: 356.0
410 -----
410 Iteration No: 136
411 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
412 with distance: 356.0
413 -----
413 Iteration No: 137
414 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
415 with distance: 356.0
416 -----
416 Iteration No: 138
417 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
418 with distance: 356.0
419 -----
419 Iteration No: 139
420 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
```

```
420 with distance: 356.0
421 -----
422 Iteration No: 140
423 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
424 with distance: 356.0
425 -----
426 Iteration No: 141
427 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
428 with distance: 356.0
429 -----
430 Iteration No: 142
431 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
432 with distance: 356.0
433 -----
434 Iteration No: 143
435 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
436 with distance: 356.0
437 -----
438 Iteration No: 144
439 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
440 with distance: 356.0
441 -----
442 Iteration No: 145
443 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
444 with distance: 356.0
445 -----
446 Iteration No: 146
447 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
```

```
441 with distance: 356.0
442 -----
443 Iteration No: 147
444 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
445 with distance: 356.0
446 -----
447 Iteration No: 148
448 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
449 with distance: 356.0
450 -----
451 Iteration No: 149
452 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
453 with distance: 356.0
454 -----
455 Iteration No: 150
456 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
457 with distance: 356.0
458 -----
459 Iteration No: 151
460 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
461 with distance: 356.0
462 -----
463 Iteration No: 152
464 The best route so far is: [6, 4, 14, 10, 3, 5, 7, 13, 11, 2, 9, 8, 12, 1, 0]
465 with distance: 356.0
466 -----
467 Iteration No: 153
468 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
```

```
462 with distance: 348.0
463 -----
464 Iteration No: 154
465 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
466 with distance: 348.0
467 -----
467 Iteration No: 155
468 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
469 with distance: 348.0
470 -----
470 Iteration No: 156
471 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
472 with distance: 348.0
473 -----
473 Iteration No: 157
474 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
475 with distance: 348.0
476 -----
476 Iteration No: 158
477 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
478 with distance: 348.0
479 -----
479 Iteration No: 159
480 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
481 with distance: 348.0
482 -----
482 Iteration No: 160
483 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
```

```
483 with distance: 348.0
484 -----
485 Iteration No: 161
486 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
487 with distance: 348.0
488 -----
489 Iteration No: 162
490 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
491 with distance: 348.0
492 -----
493 Iteration No: 163
494 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
495 with distance: 348.0
496 -----
497 Iteration No: 164
498 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
499 with distance: 348.0
500 -----
501 Iteration No: 165
502 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
503 with distance: 348.0
504 -----
505 Iteration No: 166
506 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
507 with distance: 348.0
508 -----
509 Iteration No: 167
510 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
```

```
504 with distance: 348.0
505 -----
506 Iteration No: 168
507 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
508 with distance: 348.0
509 -----
509 Iteration No: 169
510 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
511 with distance: 348.0
512 -----
512 Iteration No: 170
513 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
514 with distance: 348.0
515 -----
515 Iteration No: 171
516 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
517 with distance: 348.0
518 -----
518 Iteration No: 172
519 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
520 with distance: 348.0
521 -----
521 Iteration No: 173
522 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
523 with distance: 348.0
524 -----
524 Iteration No: 174
525 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
```

```
525 with distance: 348.0
526 -----
527 Iteration No: 175
528 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
529 with distance: 348.0
529 -----
530 Iteration No: 176
531 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
531 with distance: 348.0
532 -----
533 Iteration No: 177
534 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
534 with distance: 348.0
535 -----
536 Iteration No: 178
537 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
537 with distance: 348.0
538 -----
539 Iteration No: 179
540 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
540 with distance: 348.0
541 -----
542 Iteration No: 180
543 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
543 with distance: 348.0
544 -----
545 Iteration No: 181
546 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
```

```
546 with distance: 348.0
547 -----
548 Iteration No: 182
549 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
550 with distance: 348.0
551 -----
551 Iteration No: 183
552 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
553 with distance: 348.0
554 -----
554 Iteration No: 184
555 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
556 with distance: 348.0
557 -----
557 Iteration No: 185
558 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
559 with distance: 348.0
560 -----
560 Iteration No: 186
561 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
562 with distance: 348.0
563 -----
563 Iteration No: 187
564 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
565 with distance: 348.0
566 -----
566 Iteration No: 188
567 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
```

```
567 with distance: 348.0
568 -----
569 Iteration No: 189
570 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
571 with distance: 348.0
572 -----
573 Iteration No: 190
574 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
575 with distance: 348.0
576 -----
577 Iteration No: 191
578 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
579 with distance: 348.0
580 -----
581 Iteration No: 193
582 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
583 with distance: 348.0
584 -----
585 Iteration No: 194
586 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
587 with distance: 348.0
588 -----
589 Iteration No: 195
590 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
```

```
588 with distance: 348.0
589 -----
590 Iteration No: 196
591 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
592 with distance: 348.0
593 -----
594 Iteration No: 197
595 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
596 with distance: 348.0
597 -----
598 Iteration No: 198
599 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
600 with distance: 348.0
601 -----
602 Iteration No: 199
603 The best route so far is: [6, 4, 8, 9, 2, 11, 13, 7, 5, 3, 10, 14, 12, 1, 0]
604 with distance: 348.0
605 -----
606 End
607
608 Process finished with exit code 0
```